RetinAI Model Experiment Catalog

A repository-wide inventory of tested architectures, model variants, metrics, and training settings

Scope. This report consolidates the checked-in source, configs, result JSON/CSV files, and historical experiment notes available on 23 Oct 2026. It covers ROP screening, ROP staging, diabetic retinopathy, glaucoma, frozen-feature experiments, and early prototypes.

F1-score. The requested “F1 square” is treated as the F1-score. The report includes precision and recall when they were recorded. “NR” means the saved project evidence does not report that metric; no missing score has been estimated.

Comparability. Results use different tasks, cohorts, image sizes, score units, and evaluation splits. Treat the tables as separate experiments. An internal CV score, a held-out hospital score, a frozen-feature bag score, and a binary screening score are not interchangeable.

At a glance

Coverage	What the repository records
Backbone families	12 architecture families, including 2 ViT-S/14 pretrained-weight variants
Experiment groups	Binary ROP sweep; 6-class staging sweep and head ablations; v3 fine-tuning; frozen-feature heads; parked DR/glaucoma; early prototypes
Main reported metrics	Accuracy, macro precision, macro recall, macro/weighted F1, ROC-AUC, QWK, sensitivity, specificity, per-class precision/recall/F1, model parameter counts and inference latency where available
Status terms	Current research, historical, retired, provisional, exploratory, or invalidated are stated beside each result

Metric key: Accuracy = fraction correct; macro precision/recall/F1 weight classes equally; weighted F1 weights by class support; ROC-AUC measures score ranking; QWK is agreement for ordered labels; sensitivity and specificity are threshold-dependent.

1. Architecture inventory

All architecture families with direct training or evaluation evidence are listed below. A parameter count is shown only when a project artifact records it.

Architecture / weights	Recorded parameters	Tasks and variants found	Evidence / status
EfficientNet-B0	4.01M in binary sweep	Early binary ROP script; binary ROP sweep; 6-class ROP flat-softmax benchmark	Measured in 2 sweeps; early standalone metrics not retained
EfficientNet-B2	NR	Early binary ROP-only model	Training script/config found; usable evaluation metrics not retained
EfficientNet-B3	10.7M in legacy slide; unverified	Unified shared-backbone ROP + 5-class DR	Historical prototype; low validation/test results; superseded
EfficientNet-B5	~30M in legacy slide	Capacity / memory feasibility check	T4 batch-32 run reported OOM; no evaluation result retained
EfficientNetV2-S	20.18M backbone sweep; structured total 20,512,600	Binary ROP; 5-class DR; glaucoma; 6-class ROP flat, CORN structured, ablations; v3 baseline; frozen features	Most frequently tested backbone; several disease runs later parked or superseded
ResNet50	23.51M in binary sweep	Binary ROP sweep and screening checkpoint	Screening model retired after poor held-out-site specificity
MobileNetV3-Large	4.20M in binary sweep	Binary ROP architecture sweep	Measured in the 320 px / 12 epoch sweep
DenseNet121	6.96M in binary sweep	Binary ROP architecture sweep	Measured in the 320 px / 12 epoch sweep
ConvNeXt-Tiny, IN12K	27.8M in staging benchmark	6-class ROP flat-softmax benchmark; frozen feature encoder	Staging result is single-fold and provisional
DeiT3-Small	21.8M in staging benchmark	6-class ROP flat-softmax benchmark	Staging result is single-fold and provisional
CAFormer-S18	24.3M in staging benchmark	6-class ROP flat-softmax benchmark	Staging result is single-fold and provisional
RETFound-Green ViT-S/14-reg	NR	Frozen encoder heads and full fine-tuning for ROP staging	Foundation-model research arm; two seeds recorded
DINOv2 ViT-S/14-reg	NR	Frozen encoder heads; full fine-tuning; exam-level bag-loss variant	Foundation-model research arm; two seeds recorded

Backbone count: 12 families if RETFound-Green and DINOv2 are grouped as the same ViT-S/14-reg architecture; they are separate initialisation/model variants in the result tables.

Sources: models/common/architectures.py; results/sweep_rop.csv; results/rop/staging_comparison.json; results/rop/v3/v3_scores.json; junk/training.py; junk/training_v2.py; junk/PPT.md

2. ROP binary screening models

Binary sweep: 320 px input and 12 training epochs. Sweep metrics are from the recorded common test protocol; the full ResNet50 checkpoint is a separate 384 px run.

Model / run	Acc	Macro P	Macro R	Macro F1	QWK	ROC-AUC	Params	CPU / GPU ms	Notes
efficientnet_b0	0.758	NR	NR	0.757	0.525	0.867	4.01M	56.2 / 7.25	320 px / 12 epochs
efficientnet_v2_s	0.706	NR	NR	0.701	0.431	0.870	20.18M	130.1 / 16.83	320 px / 12 epochs
mobilenet_v3_large	0.744	NR	NR	0.743	0.500	0.857	4.20M	34.4 / 6.13	320 px / 12 epochs
resnet50	0.885	NR	NR	0.882	0.766	0.964	23.51M	157.2 / 7.28	320 px / 12 epochs
densenet121	0.764	NR	NR	0.764	0.537	0.887	6.96M	136.0 / 14.39	320 px / 12 epochs

Selected full ROP screening checkpoint: ResNet50

Evaluation	Accuracy	Macro precision	Macro recall	Macro F1	Weighted F1	QWK	ROC-AUC	Sensitivity / specificity
Internal held-out test, argmax; n=1,502	0.844	0.845	0.848	0.843	0.844	0.688	0.927	0.799 / 0.898
Screening threshold 0.1933, 5-view TTA	NR	NR	NR	NR	NR	NR	NR	0.990 / 0.398
Held-out hospital dev, same retired threshold	NR	NR	NR	NR	NR	NR	0.691	1.000 / 0.000

Reading the result. The 320 px sweep ResNet50 AUC (0.964) and the later 384 px checkpoint test AUC (0.927) come from different runs. The selected 0.1933 operating threshold reached high sensitivity but only 0.398 specificity on the internal test. At the held-out hospital, it flagged all 663 images, including healthy images, so this binary head was retired.

Sweep columns for precision and recall were not saved. NR is intentional. Latency is per-image CPU/GPU from the sweep; hardware-specific and not a clinical metric.

Sources: results/sweep_rop.csv; results/rop/metrics.json; results/rop/operating_point.json; results/rop/screening_external_spec.json; results/rop/screening_rebase.json; configs/rop.yaml

3. ROP 6-class staging: backbone benchmark

A historical single-fold, seed-42 flat-softmax comparison. Macro precision and macro recall below are computed as the unweighted mean of the saved per-class values for the six classes. Cross-site F1 is reported separately.

Model	Params	Accuracy	Macro P	Macro R	Macro F1	QWK	AUC	AP-ROP P / R	Cross-site macro F1
EfficientNetV2-S	20.2M	0.649	0.578	0.645	0.520	0.594	0.900	0.967 / 0.183	0.641
DeiT3-S	21.8M	0.584	0.557	0.615	0.517	0.541	0.881	0.892 / 0.174	0.507
ConvNeXt-T IN12K	27.8M	0.585	0.581	0.655	0.509	0.537	0.891	0.928 / 0.162	0.530
CAFormer-S18	24.3M	0.566	0.561	0.623	0.498	0.586	0.894	0.955 / 0.221	0.507
EfficientNet-B0	4.0M	0.617	NR	NR	0.492	NR	NR	NR / NR	0.536

Common run settings recorded for the staging sweep

All models used an ImageNet or named pretrained initialization, six class labels, a 384 px fundus input, dropout 0.3, batch size 16, AMP, strong augmentation, and a patient-grouped fold. The benchmark used one CV fold and one seed (42); EfficientNet-B0 ran 30 epochs, while the other four backbone runs ran 20 epochs. The records vary by benchmark pass on natural versus weighted sampling. The table is therefore a provisional screening comparison, not a definitive backbone ranking.

Across the single fold, macro-F1 spans 0.492-0.520. The held-out-site F1 column uses a different source and class mix; it is not directly comparable to the six-class CV column.

Sources: results/rop/staging_comparison.json; results/rop/staging_runs/staging_*.json; docs/ROP_PROJECT_STATE.md; kaggle_kernels/rop_staging_benchmark.py

4. ROP staging: head, sampler, and locked-set comparisons

EfficientNetV2-S is held constant in these head comparisons. Fold-0 component results are exploratory; the five-fold seed comparison is the stronger estimate.

Fold-0 variant	Macro F1	QWK	AP-ROP precision	AP-ROP recall	Evaluation note
Flat softmax, weighted sampler	0.515	0.603	0.948	0.191	Fold 0, seed 42
Flat softmax, natural sampling	0.563	0.580	0.974	0.237	Fold 0, seed 42
Structured: CORN + AP-ROP + site adversary	0.553	0.569	0.962	0.212	Fold 0, seed 42
Structured, site adversary off	0.545	0.648	0.984	0.265	Fold 0, seed 42
Structured + weighted sampler	0.403	0.129	0.985	0.141	Prior-shift failure arm
CORN removed; 5-stage softmax + AP-ROP branch	0.479	0.394	1.000	0.181	Unstable near end of training
AP-ROP branch removed	0.548	0.575	0.959	0.195	Taxonomy control

Matched 5-fold CV, natural sampling

Seed	Structured macro F1	Flat natural macro F1	Difference (structured - flat)	Finding
42	0.692 +/- 0.086	0.698 +/- 0.099	-0.0055	Flat slightly higher
1337	0.708 +/- 0.097	0.686 +/- 0.109	+0.0224	Structured slightly higher

Arm	Locked held-out hospital QWK / image	Locked held-out hospital QWK / eye	Score unit
Structured, seed 42	0.523 +/- 0.018	0.801 +/- 0.028	661 images / 230 eyes
Structured, seed 1337	0.509 +/- 0.016	0.791 +/- 0.013	661 images / 230 eyes
Flat natural, seed 42	0.561 +/- 0.023	0.795 +/- 0.007	661 images / 230 eyes
Flat natural, seed 1337	0.553 +/- 0.018	0.793 +/- 0.021	661 images / 230 eyes

The per-image to per-eye change is an evaluation-unit change using the same model predictions. Locked-set values are averages over five checkpoints. Only QWK is in this locked comparison table.

Sources: results/rop/staging_runs/ablations_fold0.json; docs/ROP_PROJECT_STATE.md; docs/paper/isbi2027_paper.md; FINAL_MODEL_EVIDENCE_PACKAGE/07_TEST_RESULTS/metrics_summary.csv

5. Structured EfficientNetV2-S: detailed metrics

The package labels this as the current structured staging model: ordinal CORN stage head, separate AP-ROP branch, and DANN site head. The source package distinguishes internal out-of-fold scores from the held-out hospital.

Metric	Internal CV OOF, 6,700 images	Locked hospital, served checkpoint, 661 images
Accuracy	0.786	0.546
Balanced accuracy / macro recall	0.652	0.520
Macro precision	0.697 (mean over 6 saved class precisions)	0.535
Macro F1, all six classes	0.652	0.476
Weighted F1	NR	0.502
QWK, 5-class ordinal map	0.897	0.493
QWK, legacy 6-label map	0.761	0.539
Unweighted kappa	0.655	0.355
Any-ROP ROC-AUC	0.969	0.835
Any-ROP average precision	0.966	0.922

Internal out-of-fold per-class report

Class	Support	Precision	Recall	F1-score
Normal	3837	0.929	0.951	0.940
Stage 1	577	0.376	0.499	0.429
Stage 2	959	0.559	0.631	0.593
Stage 3	620	0.758	0.713	0.735
Stage 4/5	89	0.663	0.775	0.715
AP-ROP	618	0.899	0.345	0.498

The external table here is the one served fold-0 checkpoint. Separate paper tables average five checkpoints and report locked eye-level QWK. They answer different questions and are kept separate.

Sources: FINAL_MODEL_EVIDENCE_PACKAGE/07_TEST_RESULTS/metrics_summary.csv; per_class_metrics/per_class_metrics.csv; 13_EXTERNAL_GENERALIZATION/external_metrics_per_checkpoint.csv; 01_MODEL/model_configuration.yaml

6. ROP v3 foundation-model fine-tuning

CV is patient-grouped, out-of-fold, pooled across two seeds, with 14 duplicate images excluded (n=6,686). Development-eye scores use 230 eyes and are descriptive; v3 has no locked-test result in the saved score summary.

Arm / model	CV accuracy	CV macro F1	CV QWK (5-class)	Development eye QWK / macro F1	AP-ROP recall / F1	Parameters / LR
RETFound-Green ViT-S/14-reg	0.758	0.643	0.794	0.774 / 0.563	0.268 / 0.397	NR; LR 2e-4
DINOv2 ViT-S/14-reg	0.766	0.653	0.825	0.756 / 0.566	0.389 / 0.501	NR; LR 1e-4
DINOv2 ViT-S/14-reg + exam bag loss	0.802	0.661	0.797	0.746 / 0.542	0.490 / 0.532	NR; LR 1e-4
EfficientNetV2-S flat natural baseline	0.788	0.656	0.816	0.760 / 0.573	0.382 / 0.515	20.2M backbone; LR 0.0003
Ensemble: mean probabilities of A + B + C	0.787	0.664	0.821	0.770 / 0.568	NR	ensemble; LR not trained

Recipe and model notes

RETFound-Green uses timm vit_small_patch14_reg4_dinov2 with released retina-pretrained weights; DINOv2 uses the matching .lvd142m weights. Both use a fresh six-way classifier, 392 px input, global average pooling, dropout 0.3 and drop-path 0.1. RETFound-Green uses half-range normalization; DINOv2 uses ImageNet normalization. The flat EfficientNetV2-S C arm is scored from saved predictions. Arm D changes the loss unit for Ostrava sessions: up to K=8 frames pooled with log-sum-exp, tau=1.0.

For A/B/D, pooled macro precision and macro recall are not provided in v3_scores.json; NR is used in the main results record. The displayed CV accuracy is the mean of the two seed-specific out-of-fold accuracies. CV QWK and macro-F1 are the official pooled-over-seed values.

Sources: results/rop/v3/v3_scores.json; results/rop/v3/runs/*/summary.json; configs/rop_staging_rfg.yaml; configs/rop_staging_dinov2.yaml; configs/rop_staging_dinov2_bag.yaml; docs/paper/PROTOCOL_V2.md

7. ROP frozen-feature and eye/exam aggregation tests - part 1

These were frozen image encoders with simple or MIL heads. The CV unit is a bag/examination, not a single image; the development unit is an eye. Scores must not be ranked against the fine-tuned image-level tables above.

Frozen encoder	Head / aggregation	CV bag QWK	CV macro F1	CV acc	Dev eye QWK	Dev eye macro F1
RETFound-Green, no CLAHE	Image logistic regression + max	0.823	0.672	0.744	0.629	0.384
RETFound-Green, no CLAHE	Image MLP + max	0.841	0.689	0.754	0.630	0.431
RETFound-Green, no CLAHE	Mean pooling	0.831	0.676	0.748	0.449	0.429
RETFound-Green, no CLAHE	Max pooling	0.822	0.675	0.745	0.572	0.452
RETFound-Green, no CLAHE	Gated attention pooling	0.838	0.684	0.754	0.613	0.411
DINOv2 ViT-S/14-reg, no CLAHE	Image logistic regression + max	0.816	0.670	0.739	0.705	0.449
DINOv2 ViT-S/14-reg, no CLAHE	Image MLP + max	0.835	0.693	0.751	0.691	0.460
DINOv2 ViT-S/14-reg, no CLAHE	Mean pooling	0.833	0.690	0.752	0.419	0.436
DINOv2 ViT-S/14-reg, no CLAHE	Max pooling	0.834	0.685	0.752	0.502	0.448
DINOv2 ViT-S/14-reg, no CLAHE	Gated attention pooling	0.833	0.687	0.755	0.648	0.439

Feature and head settings: RETFound-Green and DINOv2 use 392 px and 768-dimensional features; EfficientNetV2-S uses 384 px and 1,280-dimensional features. Image-level heads train on individual photos then aggregate the predicted class by maximum severity. MIL heads train with K=4 photos per bag and evaluate with up to 16. Mean, max, and gated attention pooling are separate model variants.

Sources: experiments/eye_spike/RESULTS.md; experiments/eye_spike/results.json

8. ROP frozen-feature and eye/exam aggregation tests - part 2

The RETFound-Green + CLAHE rows were an additional preprocessing ablation. The fine-tuned reference is included for context; the size-only control is a non-image baseline.

Frozen encoder / reference	Head / aggregation	CV bag QWK	CV macro F1	CV acc	Dev eye QWK	Dev eye macro F1
RETFound-Green, + CLAHE	Image logistic regression + max	0.840	0.683	0.754	0.739	0.423
RETFound-Green, + CLAHE	Image MLP + max	0.859	0.707	0.770	0.730	0.443
RETFound-Green, + CLAHE	Mean pooling	0.852	0.690	0.756	0.525	0.390
RETFound-Green, + CLAHE	Max pooling	0.841	0.689	0.758	0.611	0.340
RETFound-Green, + CLAHE	Gated attention pooling	0.854	0.702	0.767	0.651	0.372
EfficientNetV2-S structured, fine-tuned	Most severe photo reference	0.917	0.790	0.828	0.721	0.551
Bag-size one-hot control	Logistic regression on bag-size bins	0.096	0.142	0.451	0.018	0.076

The repository's written conclusion is that the attention MIL head did not beat the most-severe-photo rule. The fine-tuned reference scored higher on the pooled CV bags than the best frozen head (QWK 0.917 vs 0.859; macro F1 0.790 vs 0.707). On development eyes, QWK was similar, but macro F1 was lower for the frozen head. The experiment is explicitly exploratory: test_locked was not read.

Precision and recall were not included in the frozen-head aggregate table. CV QWK is over 6 fixed class labels; macro-F1 averages classes present in the evaluated sample.

Sources: experiments/eye_spike/RESULTS.md; experiments/eye_spike/results.json; experiments/eye_spike/ref_results.json

9. Parked cross-disease experiments: DR and glaucoma

These runs exist in the archived project history. They are included because they were tested, but their withdrawn/parked status must travel with their metrics.

Disease / evaluation	Acc	Macro P	Macro R	Macro F1	Weighted F1	QWK	AUC / endpoint	Status / note
DR 5-class, internal val (n=5,815)	0.744	0.595	0.598	0.586	0.764	0.728	0.788	EfficientNetV2-S; parked
DR 5-class, Messidor-2 external (n=1,744)	0.679	0.604	0.406	0.401	0.605	0.654	0.600	No retraining; 5-class transfer weak
DR referable, internal	NR	NR	NR	NR	NR	NR	AUC 0.906 (95% CI 0.89-0.92)	Grade >=2 endpoint; sensitivity 0.84, specificity 0.89
DR referable, Messidor-2 external	NR	NR	NR	NR	NR	NR	AUC 0.837 (95% CI 0.809-0.863)	External AUC drop 0.069
Glaucoma SMDG-only, G1020 zero-shot	0.478	0.559	0.563	0.478	0.486	0.090	0.589	AUC 0.957 internal; external failure
Glaucoma multi-source test	0.924	0.918	0.921	0.920	0.924	0.839	0.973	INVALIDATED: G1020 was double-mounted; same eyes could cross splits

DR used an EfficientNetV2-S, five grades, 384 px input, circle crop + Ben-Graham + CLAHE, pretrained weights, dropout 0.3, batch 16 with accumulation 2, 25 epochs, AdamW, LR 2e-4, weight decay 1e-4, focal loss gamma 2, class-balanced weights and weighted sampling, AMP, OneCycle, and five-view TTA. Glaucoma used the same backbone and base optimizer recipe, with optic-disc ROI enabled for the SMDG-only run; the multi-source results are not valid because G1020 images were included twice.

Glaucoma is withdrawn from scope; its high multi-source test scores must not be presented as valid performance. DR is parked; its referable endpoint is preserved as historical evidence.

Sources: junk/results/dr/metrics.json; junk/results/dr/analysis_summary.json; junk/results/external_dr.json; junk/results/glaucoma/metrics.json; junk/results/external_glaucoma.json; junk/results/glaucoma/multisource_persource.json; junk/configs/dr.yaml; junk/configs/glaucoma_multisource.yaml

10. Early prototypes and partial records

The files below show models or capacity checks that formed part of the project history. Their metric record is incomplete, so the report does not manufacture scores.

Model / task	Recorded parameters	Recorded configuration	Available result	Status
EfficientNet-B0, early standalone ROP	NR	ImageNet pretrained; 160 px; batch 16; 15 epochs; dropout 0.4; LR 5e-5; horizontal flip, rotation 15 deg, color jitter, affine; ImageNet normalization	No saved summary metrics located	Early prototype
EfficientNet-B2, standalone ROP v2	NR	ImageNet pretrained; 224 px; batch 16; 25 epochs; dropout 0.4; LR 3e-4; AdamW; WD 1e-4; mixup alpha 0.3; label smoothing 0.1; grad clip 1.0; class-weighted sampler; OneCycle 10% warm-up/cosine; patience 6	No saved summary metrics located	Early prototype
EfficientNet-B3, shared ROP + DR	10.7M reported in old slide; code prints exact count but saved value absent	Shared ImageNet backbone; two dropout-0.3 heads (ROP 2-way, DR 5-way); 224 px; weighted samplers; label smoothing 0.1; AdamW; WD 1e-4; OneCycle; seed 42	ROP test accuracy 0.733, F1 0.731, ROP recall 0.585; DR val accuracy 0.719, macro F1 0.497	Superseded
EfficientNet-B5	~30M reported in old slide	Feasibility check for unified model; batch size 32 on Kaggle T4	OOM; no trained result retained	Resource test only

Initial EfficientNet-B0 and B2 ROP scripts contain training code but no machine-readable final evaluation summaries. EfficientNet-B0 later reappears with measured sweep results in Sections 2 and 3. Those later values are not retroactively assigned to the early B0 run.

Sources: junk/training.py; junk/training_v2.py; junk/unified_model/kaggle_training.py; junk/unified_model/evaluate_multitask.py; junk/PPT.md; docs/CHALLENGES.md

11. Training and model parameter profiles

Values below are transcribed from saved run configs or as-run records. When the committed config and as-run settings disagree, the as-run value and discrepancy are shown.

Experiment profile	Architecture / input	Optimization and loss	Other material settings
Binary ROP final ResNet50	ResNet50, pretrained; 384 px; circle crop + CLAHE; dropout 0.3; binary classification	Batch 16 x grad accumulation 2; config LR 2e-4; AdamW; WD 1e-4; focal loss gamma 2; label smoothing .05; OneCycle; weighted sampler; epochs 30 in config (recorded final run stopped at epoch 12)	Strong augmentation; AMP; grad clip 1.0; TTA on; threshold .1933 selected on validation, measured with 5-view TTA; seed 42
ROP staging flat, as-run	EfficientNetV2-S; pretrained; 384 px; circle crop + letterbox + CLAHE; dropout .3; six classes	AdamW; peak LR 3e-4; WD .05; batch 16; 20 epochs; CE; label smoothing .05; class-balanced beta .999	Natural sampling (weighted_sampler=false); AMP; strong augmentation; site randomization; OneCycle 10% warm-up/cosine; no early stopping; last epoch; seeds 42 and 1337
ROP staging structured	EfficientNetV2-S; 20,512,600 total params; 384 px; 1,280-d embedding; dropout .3; all layers trainable	AdamW; peak LR 3e-4; WD .05; batch 16; 20 epochs; CORN ordinal loss over 4 logits (AP-ROP masked); AP-ROP BCE with sqrt-damped positive weight	Natural sampling; DANN site branch 1280->256->3; w_site .5, lambda ramp; fixed AP-ROP threshold .5; AMP; clip 1.0; last epoch; seeds 42 and 1337
RETFound-Green fine-tune	ViT-S/14-reg; 392 px; global average pool; fresh 6-way head; half-range normalization	AdamW; top LR 2e-4 as recorded; layer-wise decay .8; WD .05; batch 16; 20 epochs; CE; class-balanced beta .999; label smoothing .05	Dropout .3; drop-path .1; natural sampler; AMP; strong augmentation; site randomization; OneCycle; clip 1.0; no early stopping; last epoch; seeds 42 and 1337
DINOv2 fine-tune	ViT-S/14-reg .lvd142m; 392 px; global average pool; fresh 6-way head; ImageNet normalization	AdamW; LR 1e-4; layer-wise decay .8; WD .05; batch 16; 20 epochs; CE; class-balanced beta .999; label smoothing .05	Dropout .3; drop-path .1; natural sampler; AMP; strong augmentation; site randomization; OneCycle; clip 1.0; no early stopping; last epoch; seeds 42 and 1337
DINOv2 exam bag-loss arm	Same DINOv2 model as above	Same recipe; for Ostrava sessions only, K=8 frames pooled by log-sum-exp, tau=1.0, and loss computed per session	Same folds/seeds; inference remains per-image then scores by eye/exam rule
Frozen feature / MIL heads	RFG and DINO: 392 px, 768-d features; EffNetV2-S: 384 px, 1,280-d features	Features z-scored on train only; natural sampling; unweighted CE; MLP dim 128 with dropout .25; MIL heads AdamW 1e-3, WD 1e-2, cosine schedule, 30 epochs	MIL samples K=4 instances for training; up to 16 at evaluation; two seeds for Torch heads; no locked test read
DR five-class	EfficientNetV2-S; pretrained; 384 px; circle crop + Ben-Graham sigma 10 + CLAHE; dropout .3	AdamW; LR 2e-4; WD 1e-4; batch 16 x accumulation 2; 25 epochs; focal gamma 2; class-balanced + weighted sampler; OneCycle	Label smoothing .1; patience 7; AMP; grad clip 1.0; five-view TTA; referable threshold is grade >=2
Glaucoma binary	EfficientNetV2-S; pretrained; 384 px; circle crop + CLAHE; dropout .3; optic-disc ROI in SMDG-only run	AdamW; LR 2e-4; WD 1e-4; batch 16 x accumulation 2; 25 epochs; focal gamma 2; class-balanced + weighted sampler; OneCycle	Label smoothing .05; patience 7; AMP; grad clip 1.0; five-view TTA; multisource test invalidated by duplicate G1020

The binary ROP config says 30 epochs, but the result note describes a 12-epoch run. The staging structured committed YAML says weighted_sampler=true, while real matched headline runs overrode it to false. The table reports those inconsistencies rather than hiding them.

Sources: configs/rop.yaml; configs/baselines/rop_staging_flat_natural_asrun.yaml; FINAL_MODEL_EVIDENCE_PACKAGE/05_TRAINING/training_configuration.md; configs/rop_staging_structured.yaml; configs/rop_staging_rfg.yaml; configs/rop_staging_dinov2.yaml; configs/rop_staging_dinov2_bag.yaml; junk/configs/dr.yaml; junk/configs/glaucoma.yaml

12. Definitions, caveats, and source map

This closing page explains how to interpret unfilled cells and where each family of results lives.

How to read NR and mixed evaluation units

NR means that the metric was not present in the saved run artifact. A dash does not mean zero. Several tables use a threshold chosen for a target sensitivity; their accuracy/F1 are not the same as argmax accuracy/F1. Values on a development hospital used during method or threshold selection are descriptive and may be optimistic. The v3 foundation-model rows use image-level CV metrics and eye-level development metrics. Frozen-feature rows use bag-level CV metrics and eye-level development metrics.

Model status summary

Status	Models / evidence
Current ROP research model in the archived evidence package	Structured EfficientNetV2-S with CORN + AP-ROP branch + site adversary; compared against flat softmax. The latest v3 foundation-model arms are research comparisons, with CV and development results saved.
Retired	ResNet50 binary ROP screening head; external specificity collapsed to zero at its old threshold.
Parked / withdrawn	DR EfficientNetV2-S is parked. Glaucoma EfficientNetV2-S is withdrawn; the headline multi-source score is invalidated by duplicate eyes.
Provisional / exploratory	Five-backbone single-fold ROP staging sweep; frozen-feature/MIL aggregation spike; early EfficientNet prototypes and B5 feasibility check.

Primary source map

- Binary ROP: results/sweep_rop.csv; results/rop/metrics.json; results/rop/metrics_overfit.json; results/rop/screening_rebase.json.

- ROP staging backbone/head: results/rop/staging_comparison.json; results/rop/staging_runs/*.json; results/rop/staging_runs/ablations_fold0.json; docs/ROP_PROJECT_STATE.md.

- Structured-model evaluation: FINAL_MODEL_EVIDENCE_PACKAGE/05_TRAINING/training_configuration.md; 07_TEST_RESULTS/metrics_summary.csv; 07_TEST_RESULTS/per_class_metrics/per_class_metrics.csv; 13_EXTERNAL_GENERALIZATION/external_metrics_per_checkpoint.csv.

- Foundation models: results/rop/v3/v3_scores.json; results/rop/v3/runs/*/summary.json; configs/rop_staging_rfg.yaml; configs/rop_staging_dinov2.yaml; configs/rop_staging_dinov2_bag.yaml; docs/paper/PROTOCOL_V2.md.

- Frozen encoders and aggregation: experiments/eye_spike/RESULTS.md; experiments/eye_spike/results.json; experiments/eye_spike/ref_results.json.

- Parked disease models and early prototypes: junk/results/dr/*; junk/results/glaucoma/*; junk/results/external_dr.json; junk/results/external_glaucoma.json; junk/training.py; junk/training_v2.py; junk/unified_model/*; docs/CHALLENGES.md.

Prepared from the project repository on 23 Oct 2026. This is an experiment inventory, not a clinical recommendation or a claim that every score is independently validated.